Per-Eye Blink Counting from Patient Video with Detector-Guided Eye Crops and a Lightweight CNN

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Abstract

Blink rate, and whether both eyes blink together, are simple but informative signals about a patient's neuromuscular state. Counting blinks by hand from video is slow and error-prone. In this project we build an automatic tool that counts blinks separately for each eye *in a patient video.*

A pre-trained face/eye detector localises and crops each eye in every frame. A compact convolutional network that we train on 41,649 labelled eye images then classifies each crop as open or closed, and a per-eye finite-state machine turns the frame-wise predictions into discrete blink events while rejecting single-frame noise. The classifier reaches ≈99.8% validation accuracy with only 0.62 M parameters and runs in real time on a CPU. On a 41.3 s patient video the system counts ***15 blinks for the right eye and 0 for the left eye****. We analyse whether this striking asymmetry is clinical or a localisation artefact, and describe how to tell the two apart.*

1. Introduction

Why count blinks? Spontaneous blinking is easy to observe and hard to fake. Its rate and its left–right symmetry change with fatigue, dry-eye disease and several neurological conditions; a patient whose eyelid on one side closes weakly or not at all shows exactly such an asymmetry. A per-eye count from ordinary video would give clinicians a cheap, objective measurement.

Our approach. We split the problem into three stages that can each be checked on its own: (i) a *pre-trained* detector finds the eyes, so no localisation model has to be trained; (ii) a small CNN, trained on a public open/closed eye dataset, decides the state of each crop; and (iii) a state machine converts the state sequence into blinks. Running stages (ii)–(iii) independently for the two eyes gives separate counts.

Contributions.

- An end-to-end per-eye blink counter with two interchangeable eye-localisation back-ends: OpenCV Haar cascades [5] and MediaPipe FaceMesh landmarks [2, 3].
- A 618,754-parameter eye-state CNN reaching ≈99.8% validation accuracy.
- A noise-robust blink state machine and an analysis of the left/right asymmetry observed on the patient video.

2. Related Work

Landmark-based blink detection. Soukupová and Čech [4] compute the *eye aspect ratio* (EAR) from six eyelid landmarks and threshold it over time. EAR needs no training, but its threshold depends on the person and on head pose.

Appearance-based eye-state classification. An alternative is to crop the eye and classify its appearance. This is more robust to unusual eye shapes but needs a labelled dataset and a reliable crop. We follow this route and use a CNN with batch normalisation [1].

Face and eye localisation. Viola–Jones cascades [5] are fast, classical detectors that ship pre-trained with OpenCV. MediaPipe FaceMesh [2] regresses 468 3-D facial landmarks in real time, including dense eye contours.

3. Method

Figure 1 summarises the pipeline and the classifier.

3.1. Eye localisation (pre-trained)

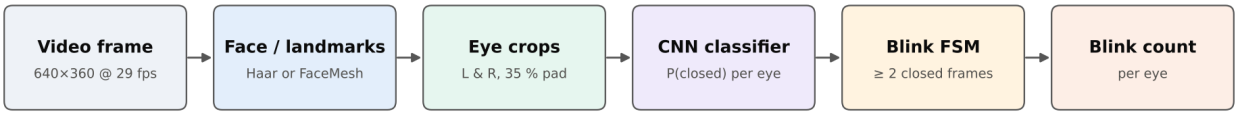
Haar back-end. The frontal-face cascade gives a face box. We search for eyes with the eye cascade only in the upper 55% of that box, split vertically at its midpoint, so that each half yields at most one eye and the two eyes can never be swapped. The largest detection in each half is kept and padded by 35% to include the eyelids, which carry most of the open/closed evidence.

Landmark back-end. FaceMesh returns 16 contour landmarks per eye; the crop is their bounding box, padded by 35% horizontally and 52.5% vertically. Landmark crops follow the eye even when it is closed, which removes the per-eye detection dropouts of the Haar back-end.

Throughout, "left" and "right" refer to the image halves as seen by the viewer.

3.2. Eye-state classifier

Each crop is resized to 50×50, replicated to three channels and normalised to $[-1, 1]$. The network (Fig. 1b, Table 1) has three convolutional blocks followed by two fully connected



(a) End-to-end pipeline, run independently for the left and right eye.



(b) Eye-state CNN: three conv blocks (32/64/128 filters) and a 2048→256→2 head.

Figure 1. System overview. Only the blink state machine carries information across frames.

Table 1. Classifier layers. Parameter counts include batch-norm and were verified by instantiating the model.

Block	Output	Params
Conv 3×3 (32), BN, ReLU, MaxPool	$32 \times 25 \times 25$	960
Conv 3×3 (64), BN, ReLU, MaxPool	$64 \times 12 \times 12$	18,624
Conv 3×3 (128), BN, ReLU, GAP	$128 \times 4 \times 4$	74,112
Linear 2048→256, ReLU, Dropout	256	524,544
Linear 256→2	2	514
Total		**618,754**

layers. Batch normalisation stabilises training, adaptive average pooling fixes the head size regardless of input size, and dropout ($p = 0.4$) regularises the head. A crop is labelled *closed* when $P(\text{closed}) > 0.5$.

3.3. Blink state machine

Frame-wise predictions are noisy, so a two-state machine (OPEN, CLOSED) runs per eye. It enters CLOSED on the first closed frame and counts a blink when the eye reopens *after at least two consecutive closed frames*; shorter closures are discarded as noise. Frames in which the eye is not found leave the state unchanged. At 29 fps two frames last 69 ms, well below the duration of a typical blink (roughly 100–400 ms), so real blinks survive while isolated misclassifications do not.

4. Experiments

4.1. Data

Classifier training data. The blink-detection dataset contains 41,649 grayscale 50×50 eye images: 19,753 closed and 21,896 open. We use a seeded random 85/15 split (35,402 training / 6,247 validation images).

Patient video. A single 640×360, 29 fps recording of 1,198 frames (41.3 s). A face is detected in every frame.

4.2. Implementation details

The classifier is trained for 15 epochs with Adam (learning rate 10^{-3}, weight decay 10^{-4}), batch size 64 and cross-entropy loss; the learning rate is halved every 5 epochs. Training images are augmented with horizontal flips, $\pm 10°$ rotations and brightness/contrast jitter. The checkpoint with the best validation accuracy is used for inference. The results below use the Haar back-end. The Haar path in the released code is a re-implementation of that pipeline, and the landmark back-end has not yet been run on the patient video.

Table 2. Per-eye results on the 41.3 s patient video (Haar back-end).

Metric	Left eye	Right eye
Blink count	**0**	**15**
Blink rate (per min)	0.0	21.8
Frames with eye detected	1,198	1,049
Frames classified closed	0	62
Frames with eye not detected	0	149

4.3. Classifier accuracy

Validation accuracy rises from 98.2% after the first epoch to ≈99.8% at the last (Fig. 3). The steps in the loss after epochs 5 and 10 coincide with the learning-rate halvings. The task is visually simple at this resolution, which is why such a small model suffices.

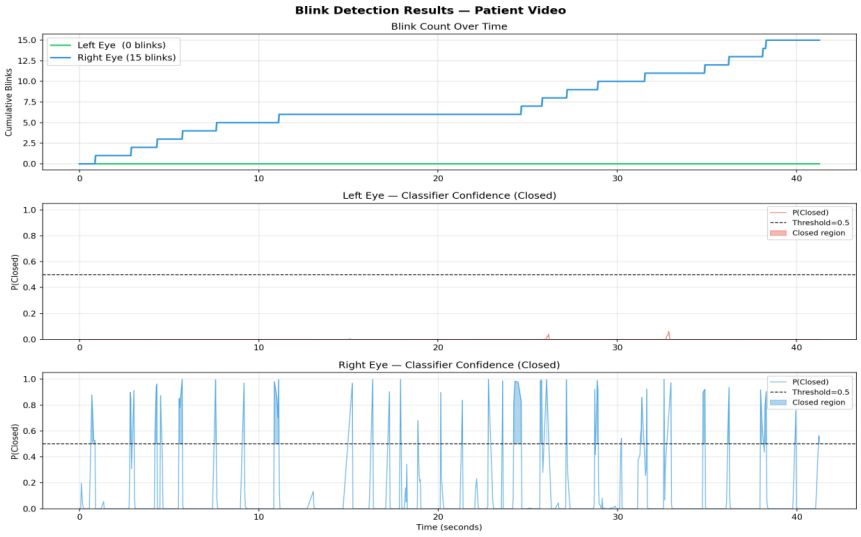
4.4. Blink counts

Table 2 and Fig. 2 give the results. The right eye produces 15 blinks (21.8 per minute, slightly above the commonly quoted resting range of about 12–20 per minute). The left eye produces none: its $P(\text{closed})$ never exceeds about 0.06.

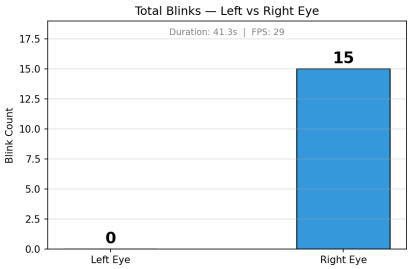
4.5. Is the asymmetry real?

Complete silence in one eye while the other blinks normally admits two explanations, and the current evidence does not separate them:

- **A genuine unilateral blink deficit.** The subject is a patient, and a one-sided inability to close the eyelid produces exactly this pattern.
- **A localisation or domain-shift failure.** The video crops (Fig. 2c) include far more eyebrow and cheek than the



(a) Cumulative blinks (top) and frame-wise $P(\text{closed})$ for each eye against the 0.5 threshold.



(b) Blink totals per eye.



(c) Eye crops from the video, all classified open; note the eyebrow and cheek inside the crop.

Figure 2. Blink counting on the 41.3 s patient video. The right eye crosses the threshold about 30 times; 15 of these closures last at least two frames and are counted. The left eye never does.

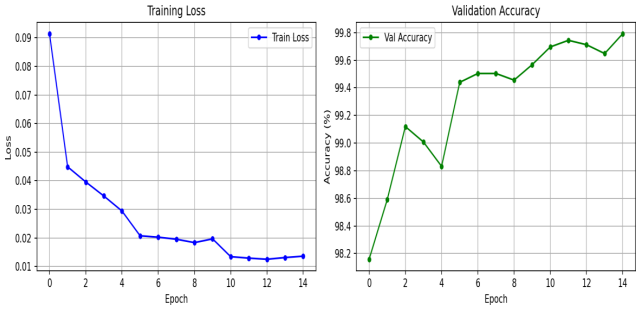


Figure 3. Classifier training loss (left) and validation accuracy (right) over 15 epochs.

tightly cropped training images. If the left crop is offset, or the cascade locks onto the eyebrow, the classifier will see "open" in every frame.

The test is direct: step through the annotated output video at a few right-eye blink times (e.g. 0.9 s, 2.9 s and 4.4 s in Fig. 2a) and check whether the left eyelid visibly moves and whether its box sits on the eye. Re-running with the landmark back-end gives an independent localisation.

5. Conclusion & Future Work

We built a per-eye blink counter from a pre-trained eye localiser, a 0.62 M-parameter CNN with ≈99.8% validation accuracy, and a simple state machine that suppresses one-frame noise. On the patient video it reports 15 right-eye and 0 left-eye blinks.

Limitations. The counts have not yet been checked against a human-annotated blink list, so precision and recall of the counter are unknown. 149 right-eye frames (12%) had no detection, and a blink inside such a gap would be missed. The 99.8% accuracy is measured on dataset images, not on video crops, so it does not automatically transfer.

Next steps. (i) Hand-annotate blink frames to measure the counter; (ii) adopt landmark crops by default and normalise their framing to match the training data; (iii) fuse the CNN probability with the EAR signal [4] for robustness; and (iv) report blink duration and inter-eye latency, which are clinically richer than counts alone.

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